# An infinite stable tournament has a transitive subtournament preserving a prescribed ordinal well-foundedness

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*Editorial note.* The counted unit is the article's own Corollary cor:sem-wf at source lines 1060-1066 together with the proof environment that begins on the next line (1067-1070); the pairing is the ordinary adjacent one.

*Editorial note.* The retained context at source lines 457-459 and 760-767 is the statement of the two results that the counted proof invokes, namely the omega-extension theorem and the main proposition of the SEM section; their proofs are not retained and they are treated as prerequisites.

*Editorial note.* The retained context at source lines 457-459 carries the author citation of Simpson and Smith inside the title of the theorem, and the author own bibliography file biblio.bib from the pinned e-print of arXiv:2402.11616v3 is attached to this task unedited and complete, with the bibtex backend, so that the citation resolves. The interval part of the pre-card gate cannot run for paper sources, so the public-interval comparison for this card was made directly against the public index: the only interval recorded there for this source is 522-524 and the retained intervals here are 457-459, 760-767, 1060-1066 and 1067-1070.

*Editorial note.* The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

*Editorial note.* The article's conservation theorems and its earlier COH, WKL and SEM lemmas are not counted here and are not used as premises beyond the two retained statements.

*Allowed standard prerequisites (editorial).*

- The SEM forcing and its main proposition. For a countable model of RCA_0 plus collection for Sigma^0_2 plus well-foundedness of the supremum of the ordinals omega to the natural product with omega below

alpha, topped by a set Y, and for every set P Turing above the jump of Y with the model of P satisfying RCA_0^*, the forcing produces a set G which is an M-infinite transitive subtournament, computes the jump of G plus Y, and whose model satisfies RCA_0 plus collection plus WF(alpha); the author's own statement is retained at source lines 760-767.. This is the statement the counted corollary specialises: the corollary drops the auxiliary set P by choosing it with the omega-extension theorem.

- Omega-extension of a model of RCA_0^* into a model of RCA_0^* plus WKL. Every countable model of RCA_0^* can be omega-extended into a model of RCA_0^* plus weak Konig's lemma; retained with this card at source lines 457-459.. The proof uses it to manufacture the set P above the jump of Y that the main proposition requires.

**Theorem 0.1** (Simpson–Smith [1])**.** *Every countable model of* $\mathsf{RCA}_0^*$ *can be* $\omega$*-extended into a model of* $\mathsf{RCA}_0^* + \mathsf{WKL}$.

**Proposition 0.2.** *Consider a countable model* $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ *topped by a set* $Y \in S$*, where* $\alpha \in M$ *is a non-zero ordinal* $\leqslant \epsilon_0$*. Then for every infinite stable tournament* $R$ *in* $S$ *and every set* $P \gg Y'$ *such that* $\mathcal{M}[P] \models \mathsf{RCA}_0^*$*, there exists* $G \subseteq M$ *such that*

1. $G$ *is an* $M$*-infinite transitive subtournament ;*
2. $P \geqslant_T (G \oplus Y)'$ *;*
3. $\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.

**Corollary 0.3.** *Consider a countable model* $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ *topped by a set* $Y \in S$*, where* $\alpha \in M$ *is an ordinal* $\leqslant \epsilon_0$*. Then for every infinite stable tournament* $R$ *in* $S$*, there exists* $G \subseteq M$ *such that*

1. $G$ *is an* $M$*-infinite transitive subtournament ;*
2. $\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.

*Proof.* By Theorem 0.1, there exists a set $P \subseteq M$ such that $P \gg Y'$ and $\mathcal{M}[P] \models \mathsf{RCA}_0^*$. The conclusion follows from Proposition 0.2. □

# References


[1] Stephen G. Simpson and Rick L. Smith. Factorization of polynomials and $\Sigma^0_1$ induction. volume 31, pages 289–306. 1986. Special issue: second Southeast Asian logic conference (Bangkok, 1984).